

A Lightweight MobileNetV2 with Dual-Level Feature Fusion for Fruit and Vegetables Freshness Classification

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Abstract—Automated produce freshness classification can help in reducing the dependency on manual inspection in quality control. The majority of lightweight CNNs use only the final backbone representations for classification while ignoring mid-level texture representations related to produce degradation. In this paper, we present MobileNetV2-Fusion that fuses an intermediate representation generated by the block_13_project_BN layer with the final out_relu representation of the MobileNetV2 architecture. For testing our approach, five different architectures, namely, VGG16, EfficientNetB0, MobileNetV2, MobileNetV2-Fusion, and MobileViT-XS, were evaluated through a matched two-stage transfer learning protocol using the balanced class weights. MobileNetV2-Fusion, which consists of 2,653,022 parameters (+2.09% more than MobileNetV2), achieved the best test accuracy (99.19%), the lowest test loss (0.0297), and the highest Weighted-F1 (0.9919), whereas EfficientNetB0 got the highest Macro-F1 (0.9854).

Keywords—produce freshness classification, MobileNetV2, dual-level feature fusion, lightweight convolutional neural network, transfer learning.

I. INTRODUCTION

Food safety and quality assurance are persistent challenges across the agricultural supply chain, where spoiled produce poses health risks and premature discarding increases food loss [1]. Manual visual inspection remains dominant but is subjective, labor-intensive, and error-prone, motivating scalable computer-vision monitoring systems for sorting, retail, and mobile use [1], [2].

However, CNNs provide automatic hierarchical feature extraction and work well for agricultural problems like crop disease detection and quality grading [3]. High-capacity models like VGG16 [4] and EfficientNetB0 [5] reach high accuracies; however, they are not appropriate for use in edge scenarios due to the large number of parameters, huge memory needs, and inference latency. Lightweight backbones like MobileNet [6] and MobileNetV2 [7] try to handle this trade-off by using depthwise separable convolutions and inverted residual bottlenecks but classify from only one of the

last backbone layers, thereby collapsing spatial information into high-level semantics and ignoring mid-level texture information which can be helpful in detecting early degeneration. Multi-level feature fusion helps overcome this problem in object detection [8] and in fusion of deep features with texture features for agriculture problems [9].

Fine-grained inter-class discrimination has been attempted via multi-task learning for quality attributes [10], densely connected networks which allow cross-layer feature re-use [11], and using fully-connected or residual layers in wide and high-capacity networks [4], [12]. However, even though visually different fruits or vegetables (e.g., rotting potatoes and guavas) exhibit the same characteristics of color change, fungal growth, and texture degradation, discriminating them based on global features alone becomes very difficult without high latency and memory costs [10]–[12]. Recently, hybrid CNN-Transformer architecture models like MobileViT-XS [13] have come up as another efficient option.

In this paper, we attempt to address this trade-off efficiently through MobileNetV2-Fusion, where MobileNetV2 [7] is extended with dual-level feature fusion, allowing capture of mid-level texture features and high-level semantic information in a unified embedding space. We evaluate MobileNetV2-Fusion against VGG16 [4], EfficientNetB0 [5], standard MobileNetV2 [7], and MobileViT-XS [13] in a 30-class, 49,156-image recognition task using a two-stage transfer-learning protocol.

II. Related works

A. CNN-Based Fruit Freshness and Quality Classification

Automated produce sorting has transitioned from handcrafted color, shape, and texture descriptors [14] to hierarchical deep CNNs [15], [16]. Amin et al. [1] demonstrated the efficacy of transfer learning for fresh/rotten classification under limited training samples. Rahman et al. [17] customized CNN backbones for produce decay grading, Ashik et al. [18] analyzed deep classifiers across multi-commodity decay stages, and Mukhiddinov et al. [19] introduced generalized freshness detection frameworks. Comprehensive overviews by Rizzo et al. [15] and Espinoza

et al. [16] systematically summarize deep learning advancements in post-harvest quality assessment.

B. Transfer Learning and Lightweight Architectures for Fruit Classification

Lightweight transfer learning backbones are vital for edge agricultural systems. Kaur et al. [20] demonstrated MobileNetV2's efficacy for produce categorization with low compute cost. Chen et al. [21] and Shahi et al. [22] introduced modified MobileNetV2 variants for fruit grading, while Gulzar [23] confirmed that MobileNetV2 achieves parity with heavier backbones under structured transfer learning pipelines.

C. Multi-Level Feature Fusion

Multi-scale feature fusion can increase discrimination between classes without adding depth to the neural network. Hierarchical feature fusion was pioneered by Feature Pyramid Networks [8]. For agricultural image processing, Hosny et al. [9] integrated CNN features with local binary pattern representations in their work on plant disease detection, while Karthik et al. [24] employed dual attention-based Swin Transformers for multi-scale symptom identification. Xue et al. [25] proposed integrating attention layers with convolutional autoencoders, while Zhang et al. [10] introduced multi-task CNNs for simultaneous quality sorting. While dense [11] and residual [12] architectures promote reusability of features, they have high memory requirements.

D. Lightweight Models for Edge and Mobile Deployment

Edge-optimized deep architectures ensure low inference time and precise diagnosis [26], [27]. Lightweight models for automation in logistics were promoted by Chen et al. [21]. Hybrid CNN-Transformer architectures like MobileViT [13] use separable convolutions along with lightweight self-attention to provide compact global-local vision, which makes this architecture a reasonable benchmark for lightweight classifiers.

II. RESEARCH METHODOLOGY

A. Dataset and Data Distribution

The current study is based on a filtered portion of the public "Freshness44" Kaggle dataset. To avoid any issues with data leakage and duplicated visuals, 14 ambiguous classes were omitted. The new dataset includes 30 equally balanced classes with 15 fruits and vegetables that can be divided into "Fresh" and "Rotten" states: apples, bananas, bell peppers, carrots, cucumbers, grapes, guavas, jujubes, limes, mangoes, oranges, pomegranates, potatoes, strawberries, and tomatoes, 49,156 images in total.

The training, validation, and test sets consist of 80%, 10%, and 10% of the total images respectively. Hence, consisting of 39,316, 4,904, and 4,936 images. In order to avoid the problem of class imbalance in the distribution of labels, class balance has been calculated based on the training set and used in both training phases. Finally, all the images were made of size 224×224 . Information about the class distribution is given in Table I.

TABLE I. DATA DISTRIBUTION

Class Name	Train (80%)	Val (10%)	Test (10%)	Total Images
Apple Fresh	2,772	346	348	3,466
Apple Rotten	3,410	426	427	4,263
Banana Fresh	2,797	349	351	3,497
Banana Rotten	2,884	360	361	3,605
Bellpepper Fresh	1,307	163	164	1,634
Bellpepper Rotten	1,686	210	212	2,108
Carrot Fresh	484	60	61	605
Carrot Rotten	405	50	52	507
Cucumber Fresh	666	83	84	833
Cucumber Rotten	553	69	70	692
Grapes Fresh	160	20	20	200
Grapes Rotten	160	20	20	200
Guava Fresh	1,081	135	136	1,352
Guava Rotten	1,063	132	134	1,329
Jujube Fresh	160	20	20	200
Jujube Rotten	160	20	20	200
Lime Fresh	875	109	110	1,094
Lime Rotten	868	108	109	1,085
Mango Fresh	311	38	40	389
Mango Rotten	474	59	60	593
Orange Fresh	2,530	316	317	3,163
Orange Rotten	2,754	344	345	3,443
Pomegranate Fresh	4,912	614	614	6,140
Pomegranate Rotten	1,109	138	140	1,387
Potato Fresh	481	60	61	602
Potato Rotten	449	56	57	562
Strawberry Fresh	642	80	81	803
Strawberry Rotten	636	79	80	795
Tomato Fresh	1,524	190	191	1,905
Tomato Rotten	2,003	250	251	2,504
Total	39,316	4,904	4,936	49,156

B. Data Processing Flow

The data processing pipeline is depicted in Fig. 1. The selected dataset was divided into training, validation, and test sets in an 80:10:10 ratio. All images were re-scaled to 224×224 px with a batch size of 32, using Keras ImageDataGenerator. Data augmentation ($\pm 20^\circ$ rotation, horizontal flipping, 0.15 zoom) and shuffling (random seed 42) were performed only on the training branch, whereas validation and test sets were not augmented. The ImageNet preprocessing step of each architecture was used before the features were extracted.

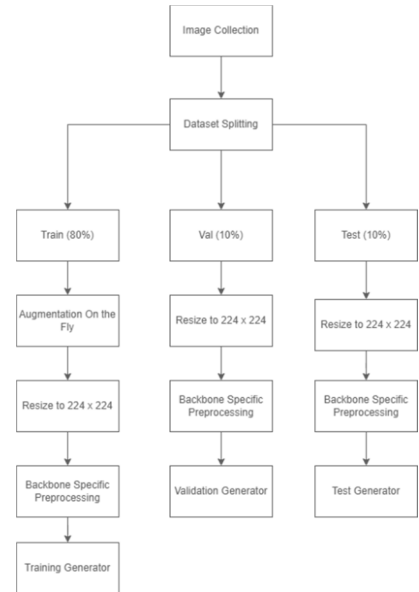


Fig 1. Diagram Process

Reproducibility was ensured by using a random seed of 42 in Python, NumPy, TensorFlow, and the batch creation process. The issue of class imbalance was solved by using balanced class weights based on the training data set while performing both warm-up and fine-tuning phases, without changing the data set but penalizing misclassification of minority classes. As illustrated in Fig. 1, online augmentation was only applied in the training generator.

C. Baseline Architectures

To provide a controlled comparison, four CNN architectures were selected as baselines, representing different design characteristics across the accuracy and efficiency spectrum.

- VGG16 [4]: a large-capacity baseline with sequential 3×3 convolutions (head is trimmed and replaced with the standard one); chosen for comparison with the fusion approach against a bigger backbone network.
- EfficientNetB0 [5]: an efficient, compound scaled baseline providing competitive performance with less number of parameters than VGG16.
- MobileNetV2 [7]: a baseline directly based on the architecture structure and employing depthwise separable convolutions, inverted residual and linear bottleneck operations; isolating the impact of the fusion branch.
- MobileViT-XS [13]: Modern hybrid CNN-Transformer model integrating separable convolutions with self-attention blocks.

All baselines were used under identical conditions in terms of data split, input resolution, batch size, data augmentation, class weighting, optimizer type, and two-step training procedure.

D. Proposed MobileNetV2-Fusion Architecture

MobileNetV2 allows efficient computations with depthwise separable convolutions, inverted residuals, and linear bottlenecks [7]; however, the baseline network classification comes from the final level of the backbone representation, missing intermediate cues about texture, color change, and spoilage. For that reason, the MobileNetV2-Fusion is proposed, which combines both intermediate and final-level representations (Fig. 2; MobileNetV2 baseline is shown in Fig. 3); it increases the number of parameters by 54,272 (2.09%) from the baseline value to 2,653,022.

The intermediate representation is extracted from block_13_project_BN ($7 \times 7 \times 160$), projected with a 1×1 convolution (128 filters), Batch Normalization, ReLU activation, and Global Average Pooling to get a 128-dimensional vector carrying localized information about textures and colors. The final representation is obtained from out_relu ($7 \times 7 \times 1280$) and averaged globally to get a 1,280-dimensional vector that carries high-level semantics of produce type and freshness.

Two vectors are concatenated to form a 1,408-dimensional fused representation, processed with Batch Normalization, 256-neuron ReLU dense layer, and Dropout (0.4), and classified with Softmax into 30 fresh/rotten classes.

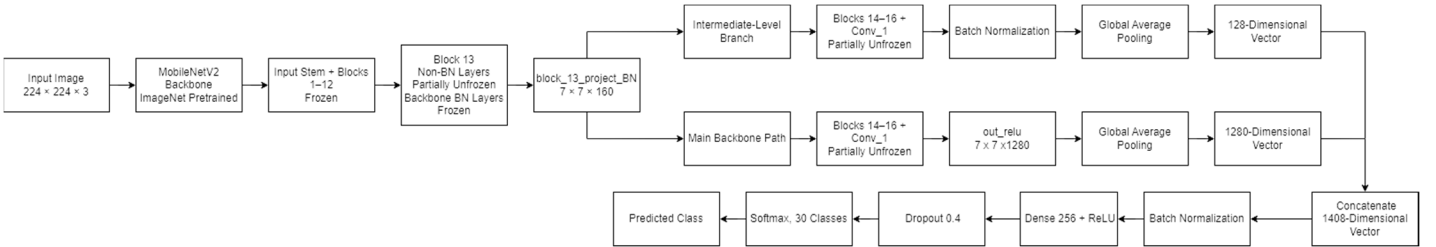


Fig 2. Architecture of MobileNetV2-Fusion Model

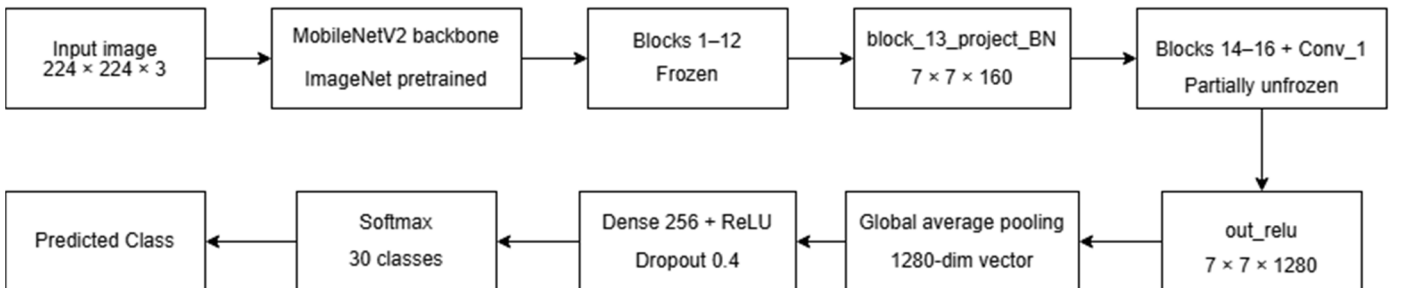


Fig 3. Architecture of MobileNetV2-Baseline Model

The proposed fusion process is based on two parallel streams:

- Intermediate level stream: The use of the $7 \times 7 \times 160$ feature map generated by the block_13_project_BN, which is mapped using an unbiased 1×1 convolutional layer that generates 128 kernels that help reduce the channels from 160 to 128 while maintaining the 7×7 spatial feature map. This is followed by Batch Normalization and ReLU activation before using Global Average Pooling to get a 128-Dimensional feature vector.
- Final-level branch: Global Average Pooling of the $7 \times 7 \times 1280$ feature map outputted by the out_relu to generate a 1280-dimensional feature vector.

E. Training Strategy and Transfer Learning

All backbone models were pretrained on ImageNet and trained using the two-stage training protocol (Table II):

- Stage 1 (Warm-up): Backbone was fully frozen, only the head model (and fusion branch when present) trained for 10 epochs, using Adam optimization algorithm, learning rate 1×10^{-4} , categorical cross entropy loss.
- Stage 2 (Fine-tuning): Selected tail layers were unfrozen (for MobileNetV2-based models, the non-BN layers in blocks 13-16 and Conv_1 with frozen BN layers, for VGG16, the block5 layer only, for EfficientNetB0, the blocks 6-7 and top_conv/top_activation with frozen BN layers), trained up to 40 epochs with Adam optimizer (learning rate 1×10^{-5}), EarlyStopping (patience 5, restoring best weights) based on validation accuracy, and ReduceLROnPlateau (factor 0.5, patience 2, minimum learning rate 1×10^{-7}) based on validation loss, while MobileViT-XS followed an analogous training schedule for consistency.

TABLE II. TWO STAGE TRAINING CONFIGURATION

Parameter	Warm-Up	Fine-Tuning
Backbone state	Fully frozen	Tail layers partially unfrozen
Optimizer	Adam	Adam
Initial learning rate	1×10^{-4}	1×10^{-5}
Loss	Categorical cross-entropy	Categorical cross-entropy
Batch size	32	32
Epochs	10	Maximum 40
ModelCheckpoint	val_accuracy	val_accuracy
EarlyStopping	-	val_accuracy, patience 5
ReduceLROnPlateau	-	val_loss, factor 0.5, patience 2
Minimum learning rate	-	1×10^{-7}
Restore best weights	Not Applied	Yes

F. Performance Evaluation Metrics

Model performance is evaluated using classification and deployment metrics consistent with prior studies [18], [20], [21]:

$$Accuracy = \frac{(TP+TN)}{(TP+TN+FP+FN)}, Precision = \frac{TP}{(TP+FP)},$$

$$Recall = \frac{TP}{(TP+FN)}, F1score = 2 \times \frac{Precision \times Recall}{Precision + Recall},$$

$$FPS = \frac{1000}{Inference\ Time\ (ms)}$$

Macro-F1 score (unweighted average) and Weighted-F1 score (weighted by class support) were both computed in the one-vs-rest setting across all 30 classes. The inference speed was measured by computing the time taken to infer on the entire test dataset comprising 4,936 images using an NVIDIA T4 GPU (batch size 32) and dividing the time by the number of test images; these measures include all the time taken for loading data, pre-processing, batching, and prediction. The FLOPs at input image size $224 \times 224 \times 3$ were calculated using the TensorFlow Profiler (for Keras models) and fvcare FlopCountAnalysis (for MobileViT-XS). McNemar's Test ($\alpha=0.05$) [32] was performed for pairwise statistical analysis of MobileNetV2-Fusion and the baselines.

III. RESULTS & DISCUSSION

A. Overall Model Performance Evaluation

The experimental results are presented in Table III. Each of the models showed high classification ability with the accuracy range between 98.70% and 99.19%.

TABLE III. EXPERIMENTAL RESULTS MAIN BENCHMARK

Model	Test Acc.	Macro F1	Weighted F1	Latency (ms)	FPS
VGG16	99.15%	0.9835	0.9915	11.51	86.88
EfficientNetB0	99.03%	0.9854	0.9902	11.24	88.95
MobileNetV2 Baseline	99.03%	0.9841	0.9903	9.44	105.88
MobileNetV2-Fusion (Proposed)	99.19%	0.9846	0.9919	11.02	90.71
MobileViT-XS	98.70%	0.9778	0.9870	12.70	78.71

VGG16 gained the second highest accuracy (99.15%) and Weighted-F1 (0.9915), but the lowest Macro-F1 (0.9835) and highest test loss (0.0415) values, which shows that the aggregate accuracy is not equal to the accuracy of the model per each class. EfficientNetB0 obtained the same level of accuracy (99.03%) as MobileNetV2 baseline, but achieved the highest Macro-F1 (0.9854), yet slightly lower Weighted-F1 (0.9902) than VGG16 and the proposed model. MobileNetV2 baseline (99.03%, 0.9841 Macro-F1, 0.9903 Weighted-F1) showed good performance, although it was slightly inferior to MobileNetV2-Fusion in all three metrics. MobileNetV2-Fusion showed the highest accuracy (99.19%), Weighted-F1 (0.9919), and the lowest test loss (0.0297) value, with the Macro-F1 (0.9846) higher than both VGG16 and MobileNetV2 but slightly lower than EfficientNetB0. MobileViT-XS scored the lowest accuracy (98.70%) among other models, meaning that additional self-attention mechanism did not help for this specific fine-grained 30-class task.

With a very small difference in accuracy of just 0.16 percentage points, McNemar's test (Table IV) was used to

establish the statistical significance of each comparison with MobileNetV2-Fusion. Each comparison was significant except the comparison with the MobileNetV2 baseline where the statistic of 1.36 was insignificant at 0.243 probability. Comparisons with EfficientNetB0 (statistic of 4765.00, $p < 0.001$), VGG16 (statistic of 4797.01, $p < 0.001$), and MobileViT-XS (statistic of 12.37, $p = 0.0004$) were statistically significant. This means that the advantage of MobileNetV2-Fusion can no longer be said to be a statistically significant one since the comparison with the mobileNetV2 baseline is not significant.

TABLE IV. PAIRWISE MCNEMAR’S TEST RESULT

Comparison	Test Statistic	p-value	Significant ($\alpha=0.05$)
MobileNetV2 vs MobileNetV2-Fusion	1.361	0.2433	No
EfficientNetB0 vs MobileNetV2-Fusion	4765.001	< 0.001	Yes
VGG16 vs MobileNetV2-Fusion	4797.005	< 0.001	Yes
MobileViT-XS vs MobileNetV2-Fusion	12.368	0.0004	Yes

B. Accuracy Efficiency Trade off

MobileNetV2 Baseline is fastest (9.44 ms/image, 105.88 FPS), which is understandable given that this network is optimized for computational speed. MobileNetV2-Fusion achieves latency of 11.02 ms/image (90.71 FPS) — which is 16.7% higher and FPS value 14.3% lower compared to MobileNetV2 Baseline — due to additional operations in its pipeline like projection, pooling, concatenation, and fusion, but still is well below 33.33 ms/frame mark required for 30 FPS real-time processing. MobileNetV2-Fusion uses 82.14% fewer parameters than VGG16 and 39.57% fewer than EfficientNetB0, outperforming both networks in terms of accuracy and Weighted-F1 score, while using just 2.09% more parameters than MobileNetV2 Baseline.

TABLE V. COMPUTATIONAL COMPLEXITY COMPARISON

Model	Parameters (M)	FLOPs (GFLOPs)	Δ FLOPs vs. MobileNetV2 (%)
MobileNetV2 (Baseline)	2.599	0.6134	—
MobileNetV2-Fusion (Proposed)	2.653	0.6155	+0.34%
MobileViT-XS	2.040	0.7232	+17.9%
EfficientNetB0	4.390	0.8015	+30.7%
VGG16	14.856	30.7132	+4907.4%

The FLOPs analysis demonstrates that the additional computation of the fusion branch is merely 0.34% compared to the MobileNetV2 baseline (0.6134 to 0.6155 GFLOPs). This is significantly lower in proportion compared to the 16.7% latency increment, thereby proving that the additional wall-clock time is caused by non-arithmetic operations (concatenation, memory read/write, and kernel launch) as opposed to increased arithmetic complexity. The MobileViT-

XS network had 17.9% FLOPs increment but performed worse in accuracy compared to all the CNN models. However, it had the longest latency time (12.70 ms) and the least FPS rate (78.71), having the fewest number of parameters (2.040M), meaning that self-attention has disproportionate overhead in terms of inference relative to the number of parameters used. The VGG16 has the most number of parameters (14.856M, +4907.4% FLOPs vs. baseline), and the highest latency among CNN models. In general, none of the models dominate every metric where EfficientNetB0 leads Macro-F1, MobileNetV2 leads Latency and throughput, and MobileNetV2-Fusion leads accuracy, Weighted-F1, and Test Loss.

C. Practical Implications and Limitations

Though the latency and FPS numbers of MobileNetV2 Fusion are good in the context of GPU-server performance, there is no direct way to understand its performance on a smartphone or an embedded device like Jetson Nano/Raspberry Pi since they have different memory bandwidth, computing capabilities, and power requirements. Deployment performance is determined based on the hardware platform, framework, numerical precision, and other optimizations (like quantization). FLOP-based analysis is useful even when GPU latency is given, because the relationship between FLOPs and latency is non-linear and platform-dependent.

IV. CONCLUSION

This study proposed a lightweight fruit freshness classification architecture based on MobileNetV2 augmented with a dual-level feature fusion mechanism. The proposed MobileNetV2-Fusion model extracts complementary feature representations from an intermediate backbone layer (block_13_project_BN) and the final backbone output (out_relu), combining them through a projection-and-concatenation fusion strategy to enrich the model's discriminative capacity.

Experimental results on a curated 49,156-image dataset spanning 30 fruit freshness classes demonstrated that MobileNetV2-Fusion achieved the highest test accuracy, lowest test loss, and highest weighted F1, while EfficientNetB0 obtained a slightly higher macro F1. The model achieved an inference time of 11.02ms per image (90.71 FPS), which demonstrates a good accuracy-efficiency balance and indicates its potential for deployment in mobile or edge systems (pending further testing and validation in operational contexts).

Further research is required to explore repeated experiments with different random seeds, external validation of the data sets, cross validation of the data sets, on device benchmarking, quantization, and pruning.

DATA AVAILABILITY

Link dataset original:

<https://www.kaggle.com/datasets/siavash93/freshness44>

Link dataset has been modified:

<https://data.mendeley.com/datasets/ns74rx364w/1>

AUTHOR CONTRIBUTORSHIP

Herlinda Angelica Tanjaya was responsible for the conceptualization of the study, development of the proposed model, implementation of the proposed MobileNetV2-Fusion architecture and experimental models, dataset preparation, and writing of the original manuscript. Kevin Reynald has been involved in the design of methodology, experiment design, validation of models, performance analysis, and revision of the manuscript. The overall research guidance, supervision, and administration have been conducted by Ribby Aulia Barlie. Guidance for methodology and technical support has been given by Irene Anindaputri Iswanto, and she did a critical appraisal throughout the entire research process. The conceptualization of the study and manuscript has been assisted by Arya Krisna Putra.

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